

SYNOPSIS

At 0938 (UTC+8) on 8 April 2005, the German registered container ship *Washington Senator*, which was on passage from Shanghai to Hong Kong at a speed of 17 knots, and the UK registered container ship *Lykes Voyager*, which was on passage from Yantian to Vancouver at a speed of 19.5 knots, collided in the Taiwan Strait. No one was hurt but both ships were damaged and, although there was no pollution, a number of containers were lost overboard. After the collision, both ships sailed to Hong Kong for repairs.

The collision occurred about 4 minutes after *Lykes Voyager* altered course to starboard from 022° to 070° to avoid *Washington Senator*, and *Washington Senator* had altered course to port from about 242° to 190° in accordance with a passing arrangement her master assumed had been made with *Lykes Voyager*. The distance between the ships at the time of the alterations was about 2.5 miles, and visibility was less than 200m in fog. Shortly after each vessel had steadied on their respective headings, both masters realised that the ships had turned towards each other, and were on a collision course. Unfortunately, by that time, the distance had further reduced to the extent that the last-minute avoiding action taken by both ships was unable to prevent a collision.

The investigation identified several contributory factors, including:

- The passing arrangement agreed by the master of *Washington Senator* was made with an unidentified ship, not *Lykes Voyager*.
- The developing close-quarters situation between *Washington Senator* and *Lykes Voyager* could have been resolved solely by the early application of the COLREGS. However, the master of *Washington Senator* opted to contact *Lykes Voyager* on VHF radio.
- By the time *Washington Senator* established VHF communications with *Lykes Voyager*, the distance between the ships was less than 5 miles.
- Identification procedures were not followed during each VHF radio transmission, and the identity of the ship with which the passing agreement was made, which was probably one of many within VHF radio range, was not established.
- The avoiding action taken by *Lykes Voyager* was not taken until the ships had closed to about 2.5 miles. This was due to the inexperience of the third officer, who was focused by the threat posed by a nearby ship being overtaken, and because the master had been distracted.
- The performance of the master of *Lykes Voyager* was possibly affected by fatigue.
- When restricted visibility and large concentrations of fishing vessels were encountered, neither the master of *Washington Senator* nor *Lykes Voyager* considered it necessary to reduce speed below their required passage speeds.

- There is no guidance available to masters of ships which are unable to proceed at a speed which allows them to be stopped within a distance appropriate to the prevailing state of visibility as required by Rule 6 of the COLREGS.
- Neither of the bridge teams made use of the available AIS information to monitor the actions of the other vessel when manoeuvring at close-quarters. This was probably due to several factors, including the absence of specific guidance or instruction from the ship managers, and the method in which AIS information was displayed.

Recommendations have been made to the Maritime and Coastguard Agency (MCA), the Federal Ministry of Transport, Building and Housing, the International Association of Marine Institutes (IAMI), the Association of Marine Educational and Training Institutes Asia-Pacific Regions (AMETIAP), and the International Chamber of Shipping (ICS) for the purpose of:

- Discouraging the use of VHF radio as a tool for collision avoidance.
- Providing additional guidance for determination of safe speed.
- Encouraging ship managers to monitor the application of safe speed.
- Highlighting the potential of AIS information when manoeuvring in a close-quarters situation.
- Advising ship managers of the advantages of displaying AIS information in a format in which it can be readily associated with radar and other sources of navigational information.
- Highlighting the importance of using ships' names and call signs in each transmission when communicating by radio.

SECTION 1 - FACTUAL INFORMATION

1.1 PARTICULARS OF *LYKES VOYAGER*, *WASHINGTON SENATOR* AND ACCIDENT

Vessel details	:	<i>Lykes Voyager</i> (photograph 1)
Registered owner	:	CPS Number 1 Ltd
Manager	:	Anglo-Eastern Ship Management
Port of registry	:	London
Flag	:	United Kingdom
Type	:	Container
Built	:	1995, Stocznia Gdynia S.A. Poland
Classification society	:	American Bureau of Shipping
Construction	:	Steel
Length overall	:	187.16m
Gross tonnage	:	23540
Engine power and/or type	:	17940kW Sulzer oil engine, direct drive
Service speed	:	19.5 knots
Persons on board	:	24
Injuries	:	None
Damage	:	Extensive damage to shell plating, decks, internal frames, and container stowages in way of the port quarter.

Vessel details:***Washington Senator* (photograph 2)**

Registered owner	:	Conti Neptun Schiffs-Beteiligungs GmbH&Co.KG MS " <i>Washington Senator</i> "
Manager	:	Niederelbe Schifffahrts-Gesellschaft (NSB)
Port of registry	:	Hamburg
Flag	:	Germany
Type	:	Container
Built	:	1994, MTW Schiffswerft GmbH - Wismar, Germany
Classification society	:	Germanischer Lloyd
Construction	:	Steel
Length overall	:	215.653m
Gross tonnage	:	34617
Engine power and/or type	:	20 510kW BV Sulzer
Service speed	:	21.3 knots
Persons on board	:	21
Injuries	:	2 (minor)
Damage	:	Extensive damage to shell plating and internal decks and bulkheads in way of the port quarter.

Accident details

Time and date	:	0938 (UTC+8) on 8 April 2005
Location of incident	:	23° 37'9N, 117° 56'9E, 069° The Brothers Light 15.4nm

1.2 NARRATIVE

(All times are UTC +8 and all courses are true)

1.2.1 *Lykes Voyager*

Lykes Voyager sailed from Yantian, China at 2006 on 7 April 2005, for passage to Vancouver with a cargo of 1800 TEU. Full ahead sea speed was selected at 2118, which gave a speed of about 18.5 knots. This was the speed required to make the ship's intended ETA in Vancouver. After writing his night orders at 2300, the master went to bed. Visibility was good, but fog was forecast for the next few days of the passage. Overnight, the master went to the bridge on two occasions after he had been informed that the visibility had reduced to about 200m, and there were large concentrations of fishing vessels in the vicinity. He remained on the bridge for between 20 and 30 minutes on each occasion. In view of the poor visibility, the ship's whistle was switched on in automatic.

After breakfast, the master returned to the bridge at about 0750. Five minutes later, the third officer relieved the chief officer as the OOW. The master remained on the bridge after the watch handover because the third officer had recently joined, and because the visibility remained at about 200m. He generally stood behind the third officer, who was using the ARPA radar on the starboard side of the bridge (**Figure 1**). The master was able to monitor both radars from this position. An AB lookout was also present. Course was 039° in autopilot to follow the planned navigational track of 038° (**Figure 2**), and the speed made good was between 19 knots and 19.5 knots. The sea was calm and the wind was light airs.

Figure 1



Photograph of the command console on the bridge of *Lykes Voyager*

Several minor course adjustments were made to avoid small radar targets, which were considered to be fishing vessels. Between 0915 and 0930, visibility increased to about 1.5nm, and a vessel being overtaken became visible on the starboard bow. The master observed this ship through binoculars. From his recollection of the letters he could see of the vessel's name, along with her general description, the vessel was identified after the accident as the Panamanian registered bulk carrier, *Notori Dake*.

At about the same time, the third officer acquired a radar target fine on the starboard bow at a range of between 8nm and 9nm. She assessed, by its vector, that it was on a course of about 235°. At 0930, the third officer altered course from 030° to 022° to increase the CPA of *Notori Dake*. For this manoeuvre, the AB was put on the helm, and manual steering was selected. Shortly after this alteration, visibility again decreased to about 200m. A call was then heard on the port VHF radio, which was set to channel 16. The call was heard only once, and the accent of the caller was judged to be Filipino or Burmese. The third officer heard a range and bearing in the transmission, but she did not hear the name of the originating ship. She was not sure if the call was intended for *Lykes Voyager*, but decided to respond.

The third officer moved to the port VHF radio, and transmitted something similar to '*this is Lykes Voyager, my position is*'. A latitude and longitude was then relayed directly from the port radar display. As soon as the third officer had finished, another ship transmitted its position, or gave a range and bearing. On hearing this transmission, which was made by a male with an Asian or eastern accent, the third officer presumed that the initial call had not been intended for *Lykes Voyager*, and returned to the starboard radar display. Neither the third officer, nor the master, recollected any further discussion on VHF radio, and assumed that the two ships had changed to another VHF channel.

During the third officer's interchange on VHF radio, the master had moved behind the chart table to read a weather facsimile. Shortly after the third officer returned to the starboard display, he resumed his position behind her. He then saw the radar target previously acquired by the third officer on the starboard bow, which had closed to 4.5 miles. This was the first time he had been aware of this target, and quickly assessed from its vector that it had a close CPA and a speed of about 23 knots. Concerned by the target's proximity and CPA, the master ordered an alteration of course to starboard to 070° to pass between the target on the starboard bow, and *Notori Dake*, which was between 1 and 1.5 miles on the starboard beam.

At the time of the alteration, in addition to the target closing on the starboard bow, and *Notori Dake*, the third officer also saw a radar target to the north west travelling at slow speed on an easterly heading. However, this target appeared to be clear of the situation that had developed. There were also a number of radar targets, which the third officer assumed to be fishing vessels, several miles distant on the port side.

Shortly before the master's intervention, the third officer had displayed the information available for the target on the starboard bow on the ARPA tote. From this, she was aware of its small CPA, and had set the radar display's EBL to 070° as a possible course for avoiding action. The master's intervention was made without discussion with the third officer, and both the master and third officer were aware that on giving the order to alter to 070°, the master had taken the con.

When steady on 070°, the CPA of the approaching target by ARPA was between 4 and 5 cables on the port side. However, about 2 minutes later, when the target was at about 5 cables, its radar vector indicated that it was turning to port towards *Lykes Voyager*. The master immediately ordered starboard 20°, and sounded an additional prolonged blast on the whistle to give warning of his proximity. At about the same time, the 'bows crossing' alert activated on the ARPA. Moments later, the bow of the approaching vessel came into view about one point off the port bow. She looked to be on a reciprocal course with a separation of about 50m, but her stern was swinging towards *Lykes Voyager*. The master ordered the helm hard to port to try and swing the stern of *Lykes Voyager* clear, but the other vessel's stern struck *Lykes Voyager* in the vicinity of No 2 crane. Contact between the two vessels was maintained for several seconds.

1.2.2 *Washington Senator*

Washington Senator sailed from Shanghai, China, at 2006 on 6 April for passage to Hong Kong with a cargo of 1531 TEU. At 0800 on 8 April, the third officer relieved the chief officer as the OOW on the bridge. Course was 233° in autopilot to follow the planned navigational track (**Figure 2**). Speed was 17 knots, which was the speed required to meet her scheduled ETA. The visibility was 1nm, and had been so for much of the night. The master had not been informed of the reduced visibility. At about 0900, the visibility further decreased to 0.5 mile, and the third officer turned the ship's whistle to automatic. The master, who was on the deck at the time, saw that visibility had reduced, and made his way to the bridge.

The master arrived on the bridge at about 0905. He initially sat at the port ARPA radar display, and the third officer sat at the starboard ARPA radar display (**Figure 3**). Visibility then reduced to less than 200m. The ship was slowly overtaking a radar target on her port bow, which was on a similar, or only a slightly diverging, course to *Washington Senator*. Its CPA by ARPA was about 9 cables within the next 10 and 12 minutes. Before the master's arrival, the AB lookout had heard a single fog signal on the port side, which had been associated with this target. The third officer saw from the AIS display that this ship was bound for Singapore; he did not note her name.



Photograph of the bridge on *Washington Senator*

A second radar target was ahead or very fine off the starboard bow. This target was moving slowly to the south, and was considered to be a fishing vessel. Its CPA by ARPA was about 2 cables. The third officer intended to pass between the two radar targets, but was told by the master to alter course to starboard and leave both vessels to port. When the third officer received this instruction, he assumed that the master had taken the con. Course was adjusted to between 240° and 243° using the autopilot. The distances of the fishing vessel ahead and the vessel on the port side when this action was taken are not known.

At about the same time as the alteration of course, the third officer informed the master of two distant radar targets on the port bow. The master monitored these targets on the port display, and instructed the third officer to also monitor them on the starboard display. After several minutes, the targets had closed to between 7nm and 8nm, and both ARPA displays indicated that the left hand of the two targets was passing clear to port. The right hand target, however, had a CPA of 2.5 cables ahead, and a speed of about 17 knots.

After being instructed by the master to obtain more information about the right hand target from the AIS display, the third officer associated the target with the AIS data shown for *Lykes Voyager*. No AIS information was obtained regarding the left hand target, which was later identified as *Notori Dake*. The master then instructed the third officer to call *Lykes Voyager* on VHF radio channel 16. The third officer recollects making three separate calls when trying to establish VHF

communication, with about a 1 to 2 minute interval between each call. In each call, the third officer broadcast information taken directly from the ARPA display. This was similar to *"ship in position (latitude and longitude), bearing, distance, course, speed, CPA, this is Washington Senator"*. The name or call sign of *Lykes Voyager* was not used, and each transmission lasted about 40 seconds. The master recollects only one call being made.

A female operator responded to the call, or calls. The third officer and master cannot recollect the exact wording of her reply, but it was similar to: *"this is Lykes Voyager (followed by a latitude and longitude and a range and bearing)"*. At the end of this transmission, the third officer repeated the ARPA information, which ended with the detail of the CPA, namely 2.5 cables. This was acknowledged by the female voice from *Lykes Voyager*. During the transmissions from *Lykes Voyager*, the master recollects hearing a male voice in the background. Shortly after the female operator had finished talking on the final occasion, the VHF transmission seemed to be immediately resumed by the same male voice, which requested *"can we pass starboard to starboard?"*

The master and third officer's accounts differ regarding whether or not a ship's name or call sign preceded this request. The third officer did not recollect a ship's name being used, but the master considered the name *Lykes Voyager* was included in the transmission, although he was not certain. However, he assumed that the request had been made by the master of *Lykes Voyager* because it had followed immediately after the female voice, and the passing arrangement requested was consistent with his own assessment that *Lykes Voyager* was impeded from altering course to starboard by *Notori Dake*. Having assessed that the action requested was safe, and after putting the AB lookout on the helm and ordering him to change to hand steering, the master agreed to the starboard to starboard passing. He also stated that he would alter 40° to port. This intended action was acknowledged by the male voice, which was again assumed to be *Lykes Voyager's* master.

The master then gave a port helm order to the AB lookout, and informed the third officer that he had taken the con. The master recalls taking this action at about 0930. The third officer moved from the starboard radar display, to check that the change from automatic to manual steering had been completed correctly, and then moved to the chart table, where he plotted the ship's position by GPS at 0935 on the paper chart. The third officer recollects that the ship was turning to port when he did this.

The ship was initially steadied on a course of 200°, which the master estimated to be the reciprocal course to that of *Lykes Voyager*. This action resulted in the CPA of *Lykes Voyager* by ARPA increasing to over 8 cables on the starboard side. However, soon after course was further adjusted to 190°, the master saw from the radar return of *Lykes Voyager's* wake, that she was altering course to starboard. Her bearing also became steady by radar. The master felt that he

was unable to alter any further to port because of *Notori Duke*, which was about 4 points off his port bow. Therefore, he ordered the helm to be put hard to starboard. Shortly after, *Lykes Voyager* became visible off the port bow. Although the helm was immediately put hard to port to try and swing the stern clear, this action was unsuccessful.

1.2.3 *Notori Duke*

Notori Duke, a 17999grt Panamanian bulk carrier, managed by Wallem Ship Management Ltd, was on passage from Hong Kong to Nakhodka, Russia. The Ukrainian third officer was the OOW, and he was accompanied on the bridge by the Russian master, a helmsman and a lookout. The ship was in hand steering and was at full ahead sea speed, making good between 15 and 16 knots over the ground. The planned track was 049° (**Figure 2**), but because of a large number of small radar targets, assumed to be fishing vessels, the ship frequently had to alter course, and generally had to keep to the east of the intended track.

The ship's two radar displays were switched on and were operating correctly. The port display, which was interfaced to a separate ARPA unit, was in true motion, fixed origin on the 3-mile range scale, and was centred. The starboard display was also in true motion, fixed origin, but was on the 6-mile range scale, and was offset to the south-west to scan about 9 miles ahead of the ship. The bridge was fitted with two VHF radios, both of which were set to channel 16, and an AIS display.

The master and third officer stated that they did not see, plot or acquire the radar targets of *Lykes Voyager* or *Washington Senator*, or observe any details relating to these ships on the AIS display. They were not aware that *Lykes Voyager* had been overtaking *Notori Duke*, and did not know that *Lykes Voyager* and *Washington Senator* had collided until 13 April. The master was aware of numerous fishing vessels in the vicinity at the time of the collision, many of which he considered to be pair trawlers, which, in his experience, frequently merge into a single target. The master felt that the presence of these trawlers might have prevented *Notori Duke* from detecting and identifying either *Lykes Voyager* or *Washington Senator*. The master and the third officer stated that they did not make, or respond to, any VHF calls during the morning of 8 April, and did not hear any "Mayday" or other emergency calls on VHF channel 16.

1.3 ACTION FOLLOWING THE COLLISION

Immediately following the collision, both masters reduced speed. The engine of *Lykes Voyager* was placed to stop, while the engine of *Washington Senator* was placed to dead slow ahead. The general alarm was sounded in both ships, and both crews were mustered.

The master of *Lykes Voyager* instructed the third officer to transmit a "Mayday" on VHF radio channel 16. He was aware the ship had suffered extensive damage, and did not know if the hull had been breached below the waterline. The "Mayday" call was made on several occasions, and, at about 1000,

Washington Senator responded. She was the only vessel to do so. In the subsequent VHF radio conversations between the two masters, the master and third officer of *Washington Senator* considered that the voice of *Lykes Voyager's* master was identical to that of the person with whom he had made the passing agreement before the collision. However, they could not be completely sure of this.

Both ships were badly damaged in the collision, and a number of containers lost overboard. A number of the remaining container stacks were also de-stabilised, particularly on board *Lykes Voyager*. The collision also caused disruptions to the power supplies in *Lykes Voyager*, and the ingress of heavy fuel from a settling tank into the engine room on board *Washington Senator*. Details of the engineering response and damage for each ship are at **Annexes A and B**.

As soon as it was apparent that *Lykes Voyager* was not in imminent danger, the "Mayday" transmissions on VHF channel 16 were stopped, and replaced with advisory messages informing other ships in the vicinity that the ship was not under command. During the period the "Mayday" and advisory messages were being transmitted by the female third officer, several cat calls, wolf whistles and lewd comments were heard on the same channel.

Washington Senator was released from the scene at 1036 and resumed her passage to Hong Kong, where she arrived early the following morning. Supported by technical advice from the ship manager, work continued on board *Lykes Voyager* throughout the day, to stabilise the containers on deck and to restore normal power supplies. The ship was able to resume passage at 2116, turning back to Hong Kong, where she anchored at 2130 the following evening.

1.4 RECORDED INFORMATION AND TRACK RECONSTRUCTION

An extract of the course recording of *Lykes Voyager*, from 0800 to 1000 on 8 April, is at **Annex C**. According to the ship's engine data logger, the time at which the master of *Lykes Voyager* stopped the engine following the collision was recorded as 0937 and 57 seconds.

Washington Senator did not carry a course recorder, and its engine data logger was not working.

The reconstructed ground tracks of *Lykes Voyager*, *Washington Senator*, and *Notori Dake* are shown at **Figure 4**. This reconstruction is based on the recorded information available, GPS positions recorded on the paper charts, navigational records and accounts from witnesses. The time of the collision has been based on the time recorded by *Lykes Voyager's* engine data logger, which is within 1 minute of the time noted on board *Washington Senator*.

1.10 THE USE OF VHF FOR COLLISION AVOIDANCE

1.10.1 *Lykes Voyager*

The master rarely used VHF radio, and never used it to resolve close-quarters situations. In circumstances where the use of VHF radio was unavoidable, such as when communicating with port authorities or pilots, he usually delegated this task to the OOW, rather than operate the VHF himself.

1.10.2 *Washington Senator*

The master of *Washington Senator* seldom used VHF for collision avoidance, but did so on this occasion because he did not want to alter course further to starboard due to the proximity of The Brothers (**Figures 2 and 4**), which would probably have had a high level of fishing activity in their vicinity. He also assessed by radar that *Lykes Voyager* was unable to alter course to starboard due to the ship she was overtaking. He recalled reading in the AIS operations manual that one of the benefits of AIS was to provide clarification when using VHF in collision avoidance.

The third officer of *Washington Senator* had previously seen the master use VHF radio to clarify an overtaking situation in the Malacca Strait, but recalled the use of VHF for collision avoidance being discouraged during his training in the Philippines.

1.10.3 Guidance

In January 2001, the MCA issued MGN 167 (M+F) titled 'Dangers in the Use of VHF Radio in Collision Avoidance', (**Annex E**). The summary of the MGN states:

Although the use of VHF radio may be justified on occasion in collision avoidance, the provisions of the Collision Regulations should remain uppermost, as misunderstandings can arise even where the language of communication is not a problem.

Guidance was also issued in the German NTM 01/05 (NFS 01/05) (**Annex F**) which stated:

Past experience has shown that the use of VHF radio by ships to agree manoeuvres in collision avoidance may in fact be the cause of serious risk situations, in particular due to:

- *the difficulty of clearly identifying other vessels at sea,*
- *misunderstandings caused by imprecise communication or insufficient language skills,*
- *collision avoidance manoeuvres not complying with the steering and sailing rules of the Regulations for Preventing Collisions at Sea.*

- *Besides, in close-quarters situations, valuable time may be lost in establishing contact on VHF radio, which should be better used to take early and effective action according to the Regulations for Preventing Collisions at Sea.*
- *When taking action to avoid a collision, the vessel's master should be aware of the fact that collision avoidance action using VHF involves risks and potentially serious consequences.*

The MCA also issued guidance on the use of VHF at sea in MGN 22(M+F) **(Annex G)**. With regard to discipline, the actions to be avoided included:

Calling on channel 16 for purposes other than distress, urgency and very brief safety communications when another calling channel is available;

Non-essential transmissions, e.g. needless and superfluous signals and correspondence;

Transmitting without correct identification; and

Use of offensive language.

1.11 THE USE OF AUTOMATIC IDENTIFICATION SYSTEMS

1.11.1 Guidance

AIS systems were fitted to both *Lykes Voyager* and *Washington Senator*, in accordance with the requirements of SOLAS chapter V, which required ships other than passenger ships and tankers, between 300 and 50,000 grt to have AIS fitted and operational by 31 December 2004 at the latest. The IMO produced guidelines for the installation of shipborne AIS in its IMO SN/Circ.227, and also provided guidelines for its operational use in its Resolution A.917 (22). The guidelines on its operational use included:

43. Once a ship has been detected, AIS can assist in tracking it as a target. By monitoring the information broadcast by that target, its actions can also be monitored. Changes in heading and course are, for example, immediately apparent, and many of the problems common to tracking targets by radar, namely clutter, target swap as ships pass close by and target loss following a fast manoeuvre, do not affect AIS. AIS can also assist in the identification of targets, by name or call sign, and by ship type and navigational status.

Additional operational guidance for the use of AIS on board ships was issued by the MCA in its MGN 277(M+F). Among the guidance included in this notice is:

Many shipowners have opted for the least cost AIS installation to meet the mandatory carriage requirement. By doing so, many of the benefits offered by graphic display (especially AIS on radar) are not realised with the 3-line 'Minimum Keyboard Display' (MKD).

Type of ship	Reporting Interval
Ship at anchor	3 minutes
Ship 0-14 knots	12 seconds
Ship 0-14 knots and changing course	4 seconds
Ship 14-23 knots	6 seconds
Ship 14-23 knots and changing course	2 seconds
Ship >23 knots	3 seconds
Ship >23 knots and changing course	2 seconds

Time Intervals for the Display of Dynamic AIS Information

1.11.2 Information provided

The information provided by AIS is divided into: static information related to a ship's descriptions, and includes its call sign and name; dynamic information, including position, and course and speed over the ground; voyage related information, including destination, draught, and hazardous cargoes; and short safety related messages. Static and voyage related information is transmitted every 6 minutes, or on request. The reporting interval for dynamic information is dependent on a ship's speed and whether or not it is changing course. This is shown at **Figure 7**.

1.12 GLOBAL MARITIME DISTRESS AND SAFETY SYSTEM (GMDSS)

The GMDSS was adopted by IMO in 1988, with transitional arrangements to be completed following a period of transition between February 1992 and February 1999. The distress and safety system used by most of the world's shipping until February 1992, required a morse radiotelegraphy watch on 500kHz for passenger ships, irrespective of size, and cargo ships of 1600 grt and above. It also required a radiotelephone watch on 2182kHz and VHF channel 16 on all passenger ships and all cargo ships of 300 grt and above on international voyages. This system relied primarily on the capability of a vessel in distress to alert another vessel for assistance. The weakness of this system was that calls for assistance might be unheard in remote areas, or even ignored.

The GMDSS is automated, and uses ship to shore alerting by means of both satellite and terrestrial radio. SAR authorities ashore, as well as shipping in the vicinity, receive the Digital Selective Calling (DSC) alert, when activated by a casualty. Shore based rescue authorities now have the primary role of co-ordinating assistance and rescue operations following the receipt of a distress alert.

Although the introduction of the GMDSS eliminated compulsory aural watchkeeping on 2182kHz, the requirement for an aural watch to be maintained on VHF channel 16 remains, to provide a distress alerting and communication channel for non-SOLAS vessels, and bridge to bridge communications for SOLAS ships.

Lykes Voyager and *Washington Senator* complied with the GMDSS carriage requirements for vessels of their size and area of operation. Both were capable of initiating a DSC alert via terrestrial and satellite radio.

SECTION 2 - ANALYSIS

2.1 AIM

The purpose of the analysis is to determine the contributory causes and circumstances of the accident as a basis for making recommendations to prevent similar accidents occurring in the future.

2.2 FATIGUE

The work patterns and rest periods of *Washington Senator*'s master and third officer indicate that they should have been reasonably well rested at the time of the collision. The third officer of *Lykes Voyager* had only recently joined, and she also appears to have had sufficient rest opportunities since travelling from the UK to Hong Kong only 3 days before the accident.

The master of *Lykes Voyager* had been on board for 2 months. He had not kept bridge watches during that time and, other than the passage between Hong Kong and Yantian, the ship's schedule provided sufficient time at sea to facilitate relatively undisrupted periods of sleep. However, although the master stated that he did not feel tired on the morning of 8 April 2005, he had achieved only 10.5 hours sleep in 6 separate periods in the previous 72 hours. This was due to cargo delays, restricted visibility, concentrations of fishing vessels, a need to accompany the inexperienced third officer on the bridge, and his inability to sleep during the opportunities available. Therefore, it is possible that his performance was affected by fatigue to some degree.

2.3 SEQUENCE OF EVENTS

Figure 4 shows that until about 0934, *Lykes Voyager* and *Washington Senator* were travelling towards each other, and that *Lykes Voyager* was overtaking *Notori Dake*. The bridge teams on *Lykes Voyager* and *Washington Senator* had detected the other ships at ranges in excess of 8 miles, and had acquired the respective targets on ARPA.

Washington Senator was the first to determine the close CPA of the two ships, and her master realised that the initial predicted passing distance of 2.5 cables required avoiding action to be taken. This was at about 0927, when the ships were 7 miles apart. The action he decided upon was not to initiate a manoeuvre, but to call *Lykes Voyager* on VHF radio.

The third officer on board *Lykes Voyager* heard a VHF radio call on channel 16. This occurred after the ship had altered to 022° at 0930 to increase the CPA of *Notori Dake*. At this time, the distance between the ships was less than 5 miles, and the effect of the alteration on the CPA of *Washington Senator* had not been considered. The third officer was not aware which ship had made the call, or if it was intended for *Lykes Voyager*, but decided to respond. After making her response, neither she, nor the master of *Lykes Voyager* participated in any

further VHF conversations before the collision. Therefore, during the VHF conversation which immediately followed, it is evident that *Washington Senator's* master agreed to a starboard to starboard passing arrangement with a person on board another, unidentified ship, but not *Lykes Voyager*.

The master of *Lykes Voyager* only became aware of the radar target of *Washington Senator*, when at about 4.5 miles. He assessed that the target had a close CPA, and he altered course to starboard to 070° to avoid it. This action was taken at about 0934, when the ships were about 2.5 miles apart. At about the same time, or very shortly after, *Washington Senator's* master altered course to port to 200° in accordance with the agreement he had made with - whom he assumed to be - *Lykes Voyager's* master.

The master of *Washington Senator* recollects altering to 200° at about 0930. However, this is not consistent with the third officer's recollection that the ship was turning to port when he plotted the GPS position on the chart at 0935. Also, had *Washington Senator* altered at 0930, she would have also been steady on 200° for about 2 miles before the collision. This does not correlate with the reconstruction of her ground track from 0901, and was unlikely to have resulted in *Notori Dake* being off her port bow on completion of the alteration to 200° as recalled by the master. The fact that the alteration was not evident from her ARPA vector on board *Lykes Voyager*, until just before the collision, also indicates that this alteration was taken a few minutes later than 0930.

Shortly after each vessel had steadied on their respective headings, both masters realised that the ships had turned towards each other, and were on a collision course. Unfortunately, by that time, the distance had further reduced to the extent that the last-minute avoiding action taken by both ships was unable to prevent collision.

2.4 THE USE OF VHF RADIO IN COLLISION AVOIDANCE

The use of VHF radio in collision avoidance is an area of contention between seafarers. Its use is favoured in some areas of the world, such as North America, but discouraged in others, including the UK and Germany. The instructions issued by Anglo Eastern, in this respect, were unambiguous and clearly directed its officers not to use VHF radio for this purpose. *Lykes Voyager's* master adhered to this policy. Of the deck officers on board *Washington Senator*, only the master used VHF to assist in collision avoidance, and then only occasionally.

The traffic situation on this occasion, although complicated by reduced visibility, was not unusual, and could have been resolved solely by the correct application of the COLREGS. When *Washington Senator's* master determined that the CPA of *Lykes Voyager* was 2.5 cables, compliance with Rules 8 and 19 of the regulations would have enabled him to alter course to starboard, to reduce speed, or both, in order to avoid the developing close-quarters situation ahead.

Had he altered course to starboard at that point, the manoeuvre would only have needed to be maintained until *Lykes Voyager* was clear. With a closing speed of 36 knots, and with the ships about 7 miles apart, this would have taken only 12 minutes. An alteration of course to starboard, could, therefore, have been made without encroaching on The Brothers, which were over 15 miles away, or encountering the fishing vessels reported to be at a distance of several miles to the north-west of *Lykes Voyager*, both of which the master stated that he was concerned about.

Even if an alteration of course to starboard was not possible, and therefore a close-quarters situation could not be avoided, Rule 19(e) required *Washington Senator* to reduce speed. This would have allowed more time for all of the vessels to accurately assess the situation and to determine appropriate avoiding action. However, the master of *Washington Senator* does not appear to have considered reducing speed, despite stating that he would do so if he considered it necessary. Instead, he opted to maintain his course and speed, and to contact *Lykes Voyager* via VHF radio. By the time contact was established, the ships had closed to within 5 miles.

After the third officer on board *Lykes Voyager* had completed her response on VHF radio, the identity of the operator who immediately followed her was not questioned, even though his transmission, unlike that of *Lykes Voyager's* third officer, was not prefixed with a ship's name or call sign. The master assumed he was speaking to the master of *Lykes Voyager* because he heard a male voice in the background when the female third officer was transmitting, and the same male voice appeared to quickly take over when the third officer had finished. His assumption was re-enforced by the proposed agreement, which accorded with his assessment of the situation between *Lykes Voyager* and *Notori Dake*. Unfortunately, it is evident that the master was wrong in his assumption, and the passing agreement was not made with the master of *Lykes Voyager*. Had proper identification procedures been followed, the identity of the ship with which the passing agreement was made, which was probably one of many within VHF radio range, would have been established.

The passing agreement made by *Washington Senator* required her to make an alteration of course to port. This was contrary to the requirements of Rule 19 of the COLREGS. It was not an action expected by the master of *Lykes Voyager*, who was unaware of the identity of the target on his starboard bow, and the passing agreement made by *Washington Senator*. The alteration of course to starboard by *Lykes Voyager* was in accordance with the requirements of Rules 8 and 19 of the COLREGS. Had *Washington Senator* agreed to a passing arrangement which also required her to manoeuvre to starboard in compliance with the COLREGS, the ships would have passed clear, despite the misidentification of ships on VHF radio.

The guidance provided in MGN 167 (M+F) (**Annex E**), acknowledges that the use of VHF radio might, occasionally, be justified in collision avoidance, and along with NFS 01/05 (**Annex F**), it highlights its potential dangers. These include: the loss of valuable time when relaying messages instead of complying with the COLREGS; the uncertainties in the identification of vessels; and the perils of manoeuvring contrary to the requirements of the COLREGS. This collision provides excellent illustrations of all these dangers.

2.5 BRIDGE TEAM MANAGEMENT - LYKES VOYAGER

Lykes Voyager's third officer detected and acquired *Washington Senator* at a range of between 8 and 9 miles, but no action was taken to avoid a collision until they had closed to about 2.5 miles. The third officer had been aware of this target for at least 9 minutes, but was still only preparing to take action when the master intervened. This was probably due to her inexperience. She had only been qualified since June 2004, and this was her second watch of her second contract.

It was not unexpected for her to be focused on keeping clear of *Notori Dake*, the nearest radar target. This was illustrated by the alteration of course to 022° at 0930. The action was successful in increasing the CPA of *Notori Dake*, but it was taken without consideration of its effect on the CPA of the radar target closing on her starboard bow, or the consequences of altering course to port in restricted visibility. The third officer's pre-occupation with *Notori Dake* was also demonstrated by her failure to view the ARPA information for *Washington Senator*, regarding her course, speed and CPA as soon as it became available after acquiring her target at about 8 miles. This perhaps also demonstrated the third officer's lack of appreciation of the 36 knot closing speed of the two vessels, which gave less than 13 minutes in which to take avoiding action. The third officer was also distracted by the VHF radio call from *Washington Senator*.

Lykes Voyager's master was aware of the third officer's inexperience. He had remained on the bridge with her for most of her first watch on board after sailing from Yantian and, because of the restricted visibility and fishing vessel activity, he had accompanied her on the bridge throughout her watch on the morning of the collision. In the conditions which prevailed, it would have been reasonable for the master to work alongside the third officer. This does not appear to have been the case, and he was not aware of the radar target of *Washington Senator* until it had been showing on radar for about 8 minutes. This was possibly because he was distracted by the sighting of *Notori Dake* during the brief period of increased visibility, when he took time to look at the ship through binoculars, and also when reading the weather facsimile at the chart table. It is possible, for the reasons outlined in paragraph 2.2, that these distractions resulted in the master's concentration being adversely affected due to fatigue. However, when he saw the target of *Washington Senator*, he assessed within about 3 minutes that there was a risk of collision, and his alteration to starboard was bold and in accordance with Rule 19 of the COLREGS.

2.8 THE USE OF AIS FOR COLLISION AVOIDANCE

Washington Senator and *Lykes Voyager* were fitted with AIS but, other than the association of the AIS information for *Lykes Voyager* with her radar target, AIS information was not utilised further. Had the initial predicted close-quarters situation between *Lykes Voyager* and *Washington Senator* been resolved, by the early application of the COLREGS, such additional information from AIS would have been of little requirement. However, as this was not the case, the use of the AIS information was potentially beneficial.

First, had *Washington Senator*'s third officer called *Lykes Voyager* on VHF radio, using her ship name, the potential for confusion would have been much reduced. *Lykes Voyager*'s third officer would have known that the call was intended for her, as would all other ships maintaining a listening watch. This potentially would have prevented the intervention of the unidentified ship, or, if not, it might have prevented *Lykes Voyager*'s third officer from abandoning the call as she did. At the very least, the use of the ship's name would have taken away the need to transmit all of the available ARPA information, making the transmission considerably shorter.

Second, following the establishment of VHF radio communications, *Lykes Voyager*'s third officer could have used AIS to associate the originator of the initial call, *Washington Senator*, with the relevant radar target.

Finally, when *Lykes Voyager* altered course to 070°, and *Washington Senator* altered course to 200°, these manoeuvres were not immediately apparent to the other vessel, primarily due to the lag in the ARPA system. This dynamic information would have been instantly available from AIS, which **Figure 7** shows would have updated changes in the heading for both ships every 6 seconds when on a steady course, and every 2 seconds when turning. Had either master monitored this information, the manoeuvring of the other ship would have been quickly apparent, and the action taken to avoid the collision at the last-minute could have been taken sooner. Following the collision, AIS could also have been used by *Lykes Voyager* to inform other AIS fitted ships of her disabled status, namely 'not under command'.

Although the IMO and the MCA have issued guidance regarding the use of AIS, it is evident from this accident that it was not used to its full potential. This was probably due to several factors, including the absence of guidance or instruction from the ship managers. Furthermore, as indicated in MGN 277, the method of display of the AIS information was also a probable factor. It was more difficult to associate radar targets with AIS information from the 3-line Minimum Keyboard Display (MKD) on board *Lykes Voyager*, than it was using the graphical display on board *Washington Senator*. However, it is possible that the AIS information might have been most easily accessed on both ships had it been interfaced directly onto their radar displays, a facility which is now available.

SECTION 3 - CONCLUSIONS

3.1 SAFETY ISSUES

1. It is possible that the performance of *Lykes Voyager*'s master was affected by fatigue, due to cargo delays, restricted visibility, concentrations of fishing vessels, a need to accompany the inexperienced third officer on the bridge, and his inability to sleep during the opportunities available. [2.2]
2. During the VHF conversation in which the master of *Washington Senator* agreed to a starboard to starboard passing arrangement, it is evident that the person the master was talking to was on board another, unidentified ship, not *Lykes Voyager*. [2.3]
3. Shortly after *Lykes Voyager* had altered course to starboard to avoid *Washington Senator*, and *Washington Senator* had altered course to port in accordance with the passing arrangement agreed with the unidentified ship, which the master of *Washington Senator* assumed to be *Lykes Voyager*, the ships were on a collision course. By the time the masters realised what had happened, the distance between the ships had reduced to the extent that the last-minute avoiding action taken by both ships was unable to prevent the collision. [2.3]
4. The traffic situation, although complicated by reduced visibility, was not unusual, and could have been resolved solely by the correct application of the COLREGS. However, the master of *Washington Senator* opted to contact *Lykes Voyager* on VHF radio. [2.4]
5. By the time VHF communications had been established between *Washington Senator* and *Lykes Voyager*, the ships were less than 5 miles apart. [2.4]
6. Had proper identification procedures been followed, the identity of the ship with which the passing agreement was made, which was probably one of many within VHF radio range, would have been established. [2.4]
7. This collision provides excellent illustrations of the dangers of using VHF radio for collision avoidance, particularly: the perils of manoeuvring contrary to the requirements of the COLREGS; the loss of valuable time when relaying messages instead of complying with the COLREGS; and the uncertainties in the identification of vessels. [2.4]
8. The avoiding action taken by *Lykes Voyager* was not taken until the ships had closed to about 2.5 miles. This was due to the inexperience of the third officer who focused on avoiding *Notori Dake*, and because the master had been distracted. [2.5]

9. During the period prior to the collision, the master of *Washington Senator* had not been informed of the reduced visibility as required by his orders, and between about 0910 and 0935, there was a misunderstanding between the master and third officer regarding who had the con. [2.6]
10. The masters of *Washington Senator*, *Lykes Voyager* and *Notori Dake* did not consider it necessary to reduce speed below their required passage speeds, when both restricted visibility and large concentrations of fishing vessels were encountered. In the experience of the MAIB, their decisions with regard to speed would have been made by many masters in similar situations. [2.7]
11. Further guidance is required to help masters determine a safe speed to cover factors such as the determination of a minimum acceptable CPA, and advice to vessels which are unable to proceed at a speed which allows them to be stopped within the prevailing visibility. [2.7]
12. Although AIS information would have provided near real-time warning of changes in each others' heading, this information was not used by either ship. This was probably due to several factors, including the absence of guidance or instruction from the ship managers, and the method in which the information was displayed. [2.8]
13. This accident is one of several recently investigated by the MAIB in which VHF radio channel 16 has been used as a means of distress by SOLAS vessels, instead of DSC. The "Mayday" was not received by a shore SAR authority and, although several ships were in the vicinity, only *Washington Senator* responded. [2.9]
14. Considerable doubt exists on the standard of radar and aural lookout maintained by *Notori Dake* during this period. [2.10]